

SEDAR 42 Gulf of Mexico Red Grouper

Data Workshop

Life History Working Group Recommendations

Stock Structure

Population genetics - LHW recommends a study using next-generation sequencing of single nucleotide polymorphisms to generate a genetic map that may elucidate sub-populations and refine the stock structure of red grouper.

Larval transport and connectivity - Implement a survey to identify red grouper Age 0's locations for an index of recruitment and identify nearshore habitats that provide recruitment to offshore areas.

Habitat Requirements - Given the expected high site fidelity of red grouper, an acoustic array around a harem may provide essential information about mating movements, spawning frequency and duration during the spawning season. Anecdotal information about cohort and feeding movements following spawning may guide more targeted tagging studies.

Tagging, movements, and migrations - Gulf wide tag-recapture programs using multiple techniques (dart tags, PIT tags, telemetry, gene tagging) to improve estimates for release mortality and movements among and across regions. Some emphasis concentrated on areas of little known information, the northern and western Gulf of Mexico, as well as the Florida Keys, and should include the time of year as a factor.

Age and Growth

Sources of Age data

- Conduct further review of current sampling methodologies by sector, including detailed comparison of length data from otolith samples and from more expansive port-based length sampling (via TIP, MRFSS/MRIP, SRHS; see Chih 2014a, 2014b).
- Bring increased attention to the need for strategies improving port sampling (representation of fishery sectors and random sampling)

- It is recommended that an increase in the number of trips intercepted by year by both the MRFSS/MRIP and SRHS occurs in the future and for a higher percentage of the intercepts include collecting biological samples (length, weight, and hard parts).

Reader Age Precision

- Continue exchanges of calibration otolith sets and age workshops among state and federal agencies and universities to continue improvements of data comparability and quality control.
- Continue use and development of a reference collection as a means to monitor precision between/among readers.
- Expand the current reference collections to include older age classes (> age 12).

Year Class Progressions

- Continue age structure sampling from all fishing sectors on an annual basis.

Age and Length Data

- Investigate methods to better collect age structure samples randomly and systematically from all fishing sectors.
- The recreational sector is still under sampling for biological samples (e.g., hard parts, $n < 100$ /year, all years). It is recommended that there is an increase in the number of trips intercepted by year by both the MRFSS/MRIP and SRHS and that a higher percentages of the intercepts include collecting biological samples (length, weight, and hard parts). In the past 4 years, only 166 trips (on average, 2010-2013; Table 4.8.11) intercepted included collecting biological samples (e.g., length, hard parts) by MRFSS/MRIP port agents and there was an estimated 22 million recreational trips made by recreational anglers in the more recent years (2010-2013; Table 4.8.17). Biological data collected at such a low percentage ($< 0.0001\%$), provide very minimal information regarding age and growth of red grouper. An increase in the number of fish intercepted for biological samples will increase our knowledge of the size and age structure being intercepted by recreational anglers.

Modeling Growth

- Explore growth model alternatives that includes both the non-random sampling due to minimum size restrictions (Diaz et al. 2004) and non-random sampling due to biases in over/under sampling specific length bins (Chih 2014a, 2014b).

Mortality

Gulf wide tag-recapture programs using multiple techniques (dart tags, PIT tags, telemetry, gene tagging) to improve estimates for natural, discard, and fishing mortalities.

Natural Mortality

- Continue the collection of otoliths from all fishing sectors, as well as, fishery independent surveys to monitor any changes in longevity.
- Continue to investigate age-varying M models and their appropriateness.
- LHW recommends further research into mortality rates of juvenile red grouper as they migrate from inshore to the offshore environment.

Total Mortality

- Continue the annual collection of otoliths from all fishing sectors, as well as, fishery independent surveys to monitor any changes in annual catch by age.

Discard Mortality

- Direct estimates of latent discard mortality are needed for the commercial sector for both bottom long line and vertical line gears. Apply innovative tag-recapture programs to the observed discards to estimate discard and other types of mortality.

Reproduction

Improve our understanding of the spatio-temporal aspects of the reproductive strategy. An example may be screen for a spatial- or depth dependence in male transition. Conduct surveys for metapopulation structure in demographics and reproduction (example hogfish assessment, SEDAR 2014b).

As in SEDAR12, the LHW recommends continued work to better understand and discriminate between annual asynchrony in spawning (skipped spawning) and seasonal asynchrony in spawning. Results of

aquaculture rearing trials, review of histology, and new information or metadata regarding temperature and the development and duration of oocytes and follicles may increase our understanding.

Age and Size at Maturity - Continue to monitor changes in maturation schedules – evidence of earlier maturity since Moe 1969.

Age and Size at Transition - Continue to monitor changes in transition schedules, evidence of earlier transition since Moe 1969.

Mating Systems - Utilize new approaches to characterize the mating system such as measurement of the amount of androgen across species and across size within species (Shepherd et al. 2013).

Develop full egg production model by accounting for temporal changes in batch fecundity and intensity of spawning and incorporate spawning frequency by size and/or age.

Meristic & Conversion factors

Continue to communicate the need to standardize length (natural total length, maximum total length, fork length, and standard length), weight (whole and gutted) measurements and the units (metric –e.g., millimeters, kilograms) used in collecting data among all sampling programs to minimize measurement errors.

Commercial Fisheries Working Group Recommendations

Landings

- -Improve data quality of CFLP Logbook VTR number to state trip ticket for data reconciliation.
- IFQ
- -Investigate dealer influence on IFQ allocation usage through dealer IFQ surveys.
- -Explore fishermen behavior in relation to allocation available.
- -Add CFLP Logbook VTR number to IFQ landing transaction form for data reconciliation.
- -Translate IFQ allocation activity ledger into a useable data set for daily allocation balances.
- -Add actual landing date to IFQ reporting form.
- -Improved enforcement of IFQ reporting infractions.
- -Improve real time seizure reporting from states law enforcements agents. Need vessel, species, pounds, price per pound, dealer, and enforcement agent.

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Discard

- -Most appropriate method for incorporation of IFQ data into discard estimations.

- -Most appropriate method for incorporation of IFQ data into discard size compositions.
- -Increased observer funding and coverage.
- -More representative observer coverage.
- -Assess reliability of effort data in logbook data.
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Overall

Meet with fishermen prior to data workshops to provide supplementary information relevant to fishery dependent data.

Recreational Fisheries Working Group Recommendations

No research recommendations were provided, though the group supported the recommendation made the Life History group:

The recreational sector is still under sampling for biological samples (e.g., hard parts, $n < 100$ /year, all years). It is recommended that there is an increase in the number of trips intercepted by year by both the MRFSS/MRIP and SRHS and that a higher percentages of the intercepts include collecting biological samples (length, weight, and hard parts). In the past 4 years, only 166 trips (on average, 2010-2013; Table 4.8.11) intercepted included collecting biological samples (e.g., length, hard parts) by MRFSS/MRIP port agents and there was an estimated 22 million recreational trips made by recreational anglers in the more recent years (2010-2013; Table 4.8.17). Biological data collected at such a low percentage ($< 0.0001\%$), provide very minimal information regarding age and growth of red grouper. An increase in the number of fish intercepted for biological samples will increase our knowledge of the size and age structure being intercepted by recreational anglers.

Indices of Relative Abundance Working Group Recommendation

- The IWG made note that the delta-lognormal index may not be the most appropriate distribution with some of the data presented and that alternative distributions should be considered. In addition, there is some variation in the SAS code used by the various labs to produce the indices. The recommendation is that a best practices workshop be convened to fully investigate different statistical models and produce a standard version of the appropriate programming code. Further, the use of R in place of SAS should be explored if the workshop warrants such consideration.

- As part of the proposed workshop, the approach to modeling 'success' in binomial portion of the delta models needs investigation. Currently, some labs model the 'proportion positive' rather than 'success' which can be an issue when used improperly.
- A calibration study is needed between the FWRI/NMFS video survey. The standardized reef systems are well suited for rigorous calibration studies, which could also include other sampling methods. In addition, exploration is needed for incorporating standardized video habitat covariates in the models.
- An exploration of the effects of IFQ's on the fishery dependent indices, especially the commercial handline and longline is needed. During the workshop, fishermen indicated that since the implementation of IFQ's, there has been a drastic change in fisheries behavior. There is also the possibility that dealers can directly influence this behavior. There is a need to incorporate these years into the overall time series in the most appropriate manner and to determine the means for doing so.
- The MRFSS data are clustered in the sense that some records represent individuals on the same boat (a cluster). An issue arose where the proper identifier for those clusters was not obvious in the data set. Hence, further investigation into 'party id' and what it represents in the MRFSS data is needed to accurately estimate the variability associated with the indices.
- Expansion of video surveys into Florida Bay
- Development of a YOY survey
- For reef-associated fisheries, the fishery-independent monitoring is based on known distribution of habitat. As side-scan sonar and similar activities increase the list of known habitat, there is a need to ensure that the sampling strategies for the FIM adjust appropriately and are optimized as habitat information becomes available.

Integrated Ecosystem Assessment Working Group Recommendations

Recommendation 1: Time varying natural mortality

Research is required to incorporate interannual variation in red grouper natural mortality within the assessment process. In particular, elevated mortality rates in fishes, including members of the shallow-

water grouper complex, can be caused by severe red tide events (Flaherty & Landsberg 2011). A red tide severity index (Walter et al. 2013) was previously included in the base stock assessment model for Gulf of Mexico gag grouper, which improved model fits to indices of abundance (Sagarese et al. 2014b). In the Gulf of Mexico gag grouper assessment (SEDAR 33), fluctuations in red tide mortality varied more than 10-fold through time, and were estimated to be commensurate with fishing mortality rates in several “severe” years (Sagarese et al. 2014b). Like the gag grouper SEDAR assessment, red tide severity should be considered as a source of mortality for red grouper. This recommendation requires at least four research steps.

First, length/age composition data are needed to determining lengths/ages susceptibility to red tide severity.

Collections of fish during red tide events would allow for the size/age selectivity of mortality to be determined, and might also allow for some minimum estimates of total mortality. Preliminary data were distributed by the NMFS Panama City lab containing red grouper lengths and estimated ages for 16 individuals collected from the Big Bend region during August 1st and 3rd of 2014. During plenary, various participants noted that collection of samples during the NMFS bottom longline survey was complicated by the decomposed nature of many fish encountered, which also prevented length estimates. In addition, otoliths were often difficult to recover from some specimens because they were missing anterior portions of their body.

Second, existing indices of red tide severity should be updated.

The IEA group recommends research to produce candidate indices of red tide severity and to devise scenarios based on red tide indices and methods for inclusion in the red grouper Stock Synthesis assessment model. Updating red tide indices is difficult because the original red tide indices (Walter et al. 2013) were created using SeaWiFS (operational 1998 – December 2010) satellite sensors. More recently, MODIS (Moderate Resolution Imaging Spectrometer) satellite sensors (July 2002 – present) have been used to detect and track harmful algal blooms (Stumpf et al. 2003, Hu et al. 2005). Thus, steps need to be taken to (i) calibrate SeaWiFS and MODIS satellite data during overlap periods; (ii) extend the red tide index through the present period (2014); and (iii) automate compilation of satellite data, and calculation and reporting of index values.

Third, procedures for incorporating red tide indices into Stock Synthesis should be critically evaluated.

Simulations should be conducted to evaluate the consequences of assuming constant or size specific natural mortality, when mortality actually fluctuates according to episodic temporal events. Further, approaches to incorporating environmental indices in stock assessment tuning procedures should be compared through simulated datasets to evaluate the effects of assessment model misspecification.

Fourth, the statistical properties of red tide indices should be characterized for use in simulations and assessment projections.

Evaluate whether all levels of red tide severity are equally likely in near-term future events, or whether information is contained in red tide indices that can be used to generate ‘forecast distributions’. Time series decomposition can be used to statistically characterize red tide indices (Stumpf et al. 2003). By quantifying periodicity, trends, and stochasticity, ‘forecast distributions’ may enable plausible future scenarios to be considered in assessment projections.

Recommendation 2: Index of red tide mortality derived from Ecopath with Ecosim

The IEA working group agreed that additional efforts deriving natural mortality values from the WFS Red tide Ecopath with Ecosim model would be helpful as presented for gag grouper during SEDAR 33 (Gray et al. 2013). These modeling efforts would allow red tide events to affect multiple components of the West Florida Shelf ecosystem and to assess the overall effect of red tide and predator/prey dynamics on the mortality rates of Gulf of Mexico red grouper.

Recommendation 3: Elucidating the response of red grouper to red tide events

Future modeling efforts should aim to address whether groupers move in response to red tide events or if they experience elevated natural mortality during these episodic events.

Recommendation 4: Modifications to the CMS modeling framework

Additional fisheries-independent data (e.g., PCLAB data) will be incorporated in the datasets used for habitat modeling of red grouper. This will allow us to improve the predictions made by the binomial GLMs described in SEDAR42-DW-04. Thus, we will be able to better predict the probability of presence of adult red grouper on the West Florida Shelf and, therefore, to better simulate the production of red grouper eggs over space in the CMS.

The life history working group brought up concerns regarding the aggregated use of all adult red groupers in determining the number of eggs released at red grouper spawning sites. There is evidence in red grouper that the fecundity of large adult females is considerably higher than that of small adult females. To account for this, the IEA group will use data compiled by the life history group to calculate mean age at depth for red grouper. This information will be useful to estimate the number of eggs released at each red grouper spawning site based on (1) the probability of presence of adult red grouper at that site; and (2) the relative fecundity at that site. The relative fecundity at each spawning site will be determined from: (1) the depth at that site; (2) the mean age at depth profile; and (3) the fecundity-at-age (number of eggs released during a spawning event at age) profile.

The CMS index should be extended to cover 2014 to provide insight into potential recruitment for the first year of projections.

Recommendation 5: Enhance fish kill reporting, particularly in offshore regions

Current understanding of fish killed by red tide events largely originates from the Florida Fish and Wildlife Conservation Commission and Fish and Wildlife Research Institute fish kill database, which is informed by a statewide fish kill hotline (<http://research.myfwc.com/fishkill/>). Many of the observations are based on fish that washed ashore following red tide events. Enhanced reporting of red tides, in addition to observations from offshore waters by recreational and commercial fishermen, could increase understanding of how red tide events impact offshore species. This could be achieved through the creation of a national program or increased citizen science through outreach educating fishermen and other Gulf patrons on their ability to improve fish kill reporting.

Assessment Workshop

1. Evaluate existing methods for deriving historical discard numbers and discard rates and improve methods as appropriate.
2. Develop/evaluate methods to maintain continuity of fishery-dependent indices in light of management regulations and ITQs.
3. Considering red tide is an unpredictable event, but can be a significant source of mortality, a response protocol should be developed for data collection and incorporation of the information into updated assessments.
4. The start year of this assessment is 1986. Future assessments should investigate extending the assessment model further back in time.

Review Workshop

Recommendations numbered in order of priority:

- 1) Questions were raised in workshop discussions about changes in reproductive success with age and with population concentration. Although it is known that reproductive success is mitigated by social factors, the degree and extent of mitigation is not well understood. More data are needed to better address the topic, including addressing time-varying changes within age categories. How much variation exists in size at age? Insufficient information leads to greater uncertainty and can have impacts on reference points.
- 2) The review panel raised questions throughout the evaluation of the assessment reports about the basis or bases of decisions to use the variables actually used. The recommendation is for analysts to provide a justification or rationale for the selection criteria. Review panel evaluations could be more accurately motivated if the selection criteria were better reported. Selection criteria would thereby be carried forward in the evaluations.
- 3) Paralleling the AW Report, discussions in the Review Workshop focused on initial conditions of the red grouper stock, including assessment of the stock from 1986 to 1993. Composition data was more complete from 1993, raising questions about methods to approach composition prior to that data, in effect, how to decide on what methods and procedures to use and how to establish preferences. The sense of the Review Workshop members was that procedures of ramping up were needed.
- 4) The core problem in the red grouper assessment was the data on discards. Procedures for reporting discards were not consistent across the fleets, and the fit to indices were poor, leading to major sources of uncertainty. Numerous sensitivity runs helped to reduce the lack of fit, especially up-weighting the commercial fishery dependent data, but problems remain. Discards were missing from the shark longline fishery, raising questions about the amount of resultant uncertainty. Research to address best practices in the reporting of discards is needed in regard to the red grouper stock. Especially useful, also, would be to increase the number of observers and observations among the commercial fleets.
- 5) Develop a standard protocol for ensuring that appropriate uncertainty in recruitment is applied when developing projections. Using a long-term average recruitment, as the median was used in this assessment, may lead to very different projections, and thus different management advice, compared to a lower or higher average recruitment based on a more recent time-period that may be more likely to reflect the biological/environmental reality of the stock.
- 6) Research is needed to help address questions about how fecundity can best be measured. Fecundity is a preferred measure for stock biomass and is a function of the number of eggs produced, but it has to be measured indirectly. Gonad weight can be used as a proxy, but obtaining reliable weights can be problematic, dependent in part on methods and timing of data acquisition. Research to address more accurate measurement of gonad weight and to develop protocols would be helpful.

- 7) Actual measures of individual growth are needed within age categories, as opposed to relying on common assumptions about growth rates and outcomes. Differential growth rates may occur in stock sub-structure within localized species, due to characteristics of the stock and ecosystem variables, or both. They may also affect schedules for hermaphroditic changes, impacting sex-age class composition.
- 8) Sensitivity runs to assess the impact of the 2005 red tide event on red grouper landings did not show any significant differences from the base model, including fit to discards. Documentation of the red tide mortality, however, presents methodological difficulties. Although analyses of data suggest that red tide primarily affected ages 0+, composition of the red tide kills are difficult to measure, given problems of access to the red tide zones and incomplete records of age, size, and sex in the kills. In addition, red tide events may be best considered in reference to ecosystem considerations (SEDAR42-5W-01). As environmental indices become incorporated into single stock species, criteria for inclusion have to be tested and measured, toward a goal of balanced biological and mechanistic explanation, statistical significance, and predictive performance. More research for red tide impacts on red grouper stock status is especially appropriate, given that the majority of landings are on Florida's West Continental Shelf, where high concentrations of red tide tend to be located.
- 9) Red grouper are found throughout the Gulf of Mexico and in the Atlantic from Brazil to the southeastern US. Catch levels and age composition data from Cuba, the Caribbean, and especially Mexico would allow for more complete stock assessment. Data could be obtained from the Mexican organization MEXAS.
- 10) The average age and thus size of females changing to males are known, but a more complete understanding of the conditions under which the changes occur would be helpful for assessing stock size. The number of males in relation to minimum stock size would be a useful metric for stock assessments.

a) Clearly denote research and monitoring that could improve the reliability of, and information provided by future assessments with particular emphasis on the Deepwater Horizon Oil Spill.

SEDAR 42 did not directly address the impacts of Deepwater Horizon on red grouper stock status. Earlier SEDAR Workshops, for example, SEDAR 31 (red snapper), contained discussions and research recommendations. SEDAR 42 contains analyses and recommendations relevant for events such as oil spills, however, in the attention given to the 2005 and 2014 red tide events. Oil spills can be measured as environmental events in a variety of ways, including the event as equivalent to a fleet source for fishing mortality. Ecosystem considerations (see 9 above) can also be utilized to assess impacts on stock status.

b) Provide recommendations on possible ways to improve the SEDAR process.

A topic of discussion throughout the Review Workshop was the need for more attention to

commonly used assumptions in categories of data and analyses. The rationale was to make assumptions more explicit and for consideration to be given to criteria for selection of options. The selection criteria would therefore become a part of the record as analyses move forward, and they would be subject to considerations of clarity, efficiency, and parsimony.

The three days of Review Workshop proved insufficient to enable the pre-prepared assessment to be presented and address areas of concern through developing additional work. For the workshop component of the review to be effective (i.e. developing alternatives or options where issues are identified), some additional time would need to be made available to the Assessment Team during